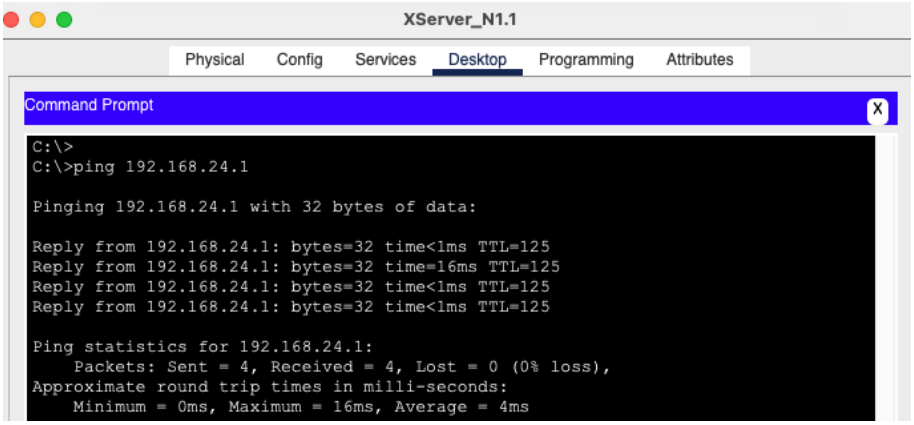

Fig.P1r_1a

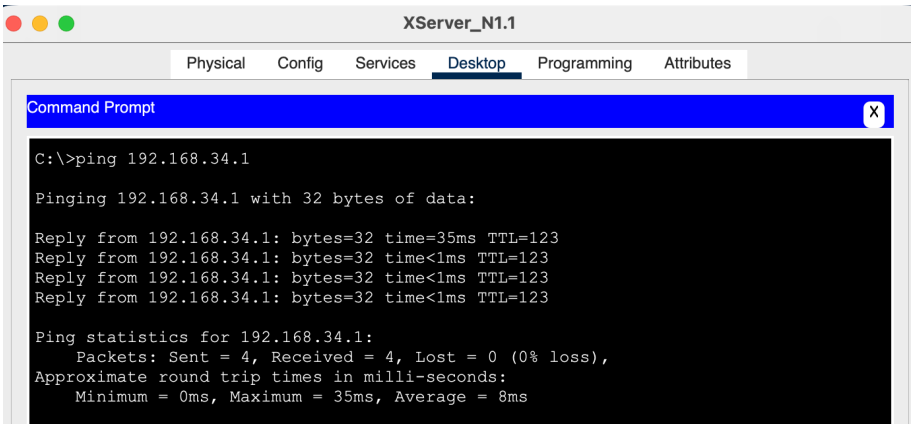

Fig.P1r_1b

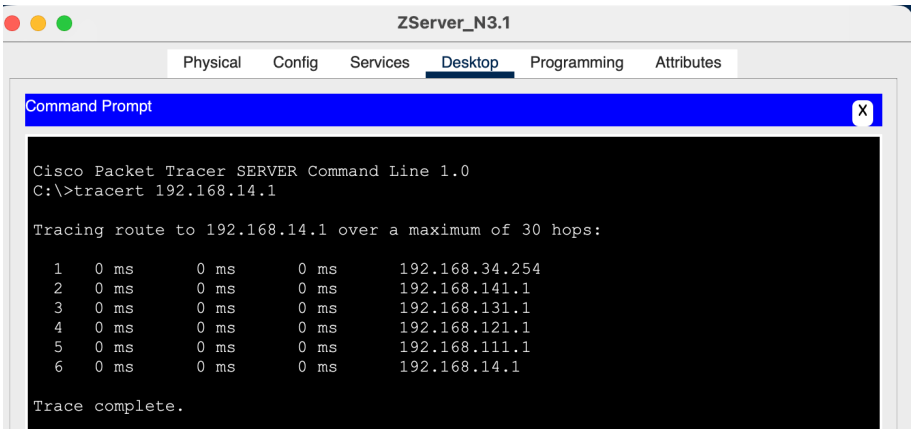

Fig.P1r_2a

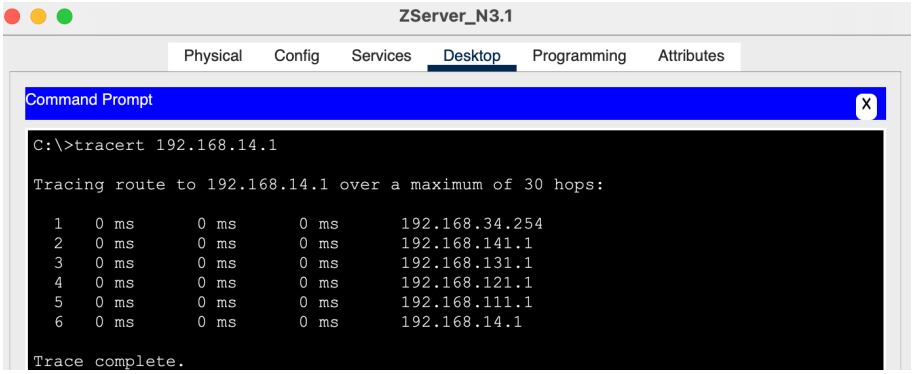

Fig.P1r_2b

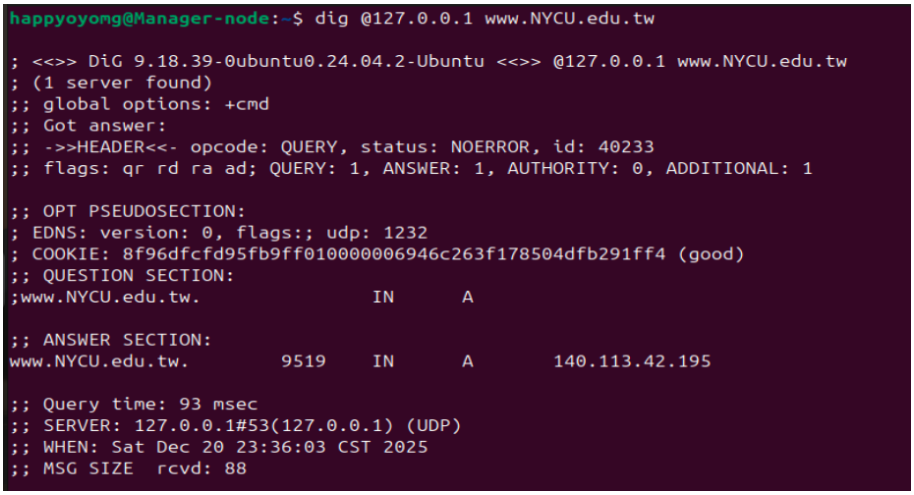

Fig.P2r_1

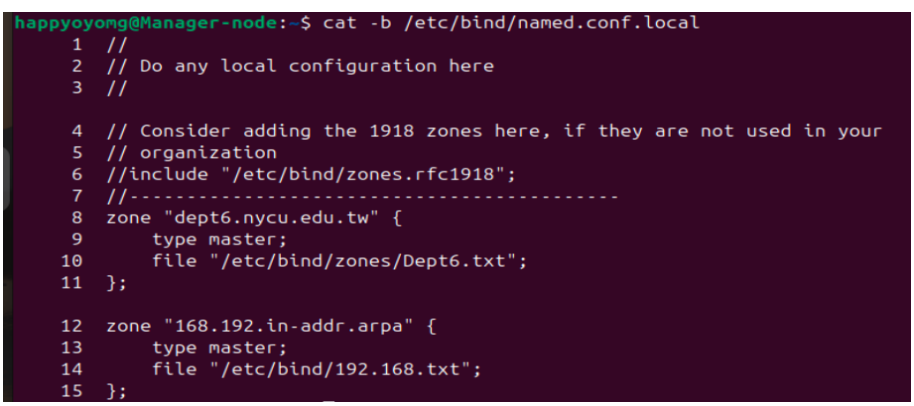

Fig.P2r_2

```
happyoyomg@Manager-node:~$ cat -b /etc/bind/db.dept6
     1  ;
     2  ; BIND data file for domain zone: dept6.nycu.edu.tw
     3  ; last revised: 2025.12.10
     4  ;

     5  $TTL    604800
     6  @       IN      SOA     localhost. root.localhost. (
     7                          2025121002      ; Serial
     8                          604800          ; Refresh
     9                          86400           ; Retry
    10                          2419200         ; Expire
    11                          604800 )        ; Negative Cache TTL

    12  ; Name Server
    13  @       IN      NS      dns.dept6.nycu.edu.tw.

    14  ; DNS server
    15  dns     IN      A       192.168.56.2     ; VM IP
    16  ;dns    IN      A       192.168.34.1      ; Packet Tracer (可留或註解)

    17  ; Hosts
    18  n1dhcp  IN      A       192.168.14.1
    19  n2dhcp  IN      A       192.168.24.1
    20  mail    IN      A       192.168.34.25
    21  www     IN      A       192.168.34.80
    22  n4dhcp  IN      A       192.168.44.1
```

Fig.P2r_3a

```
happyoyomg@Manager-node:~$ cat -b /etc/bind/db.192-168
     1  ;
     2  ; BIND reverse data file for the zone "168.192.in-addr.arpa"
     3  ;

     4  $TTL    604800
     5  @       IN      SOA     localhost. root.localhost. (
     6                          2025121001      ; Serial
     7                          604800          ; Refresh
     8                          86400           ; Retry
     9                          2419200         ; Expire
    10                          604800 )        ; Negative Cache TTL

    11  ; Name Server
    12  @       IN      NS      dns.dept6.nycu.edu.tw.

    13  ; PTR records (IP -> Hostname)
    14  1.14.168.192.in-addr.arpa.   IN  PTR  n1dhcp.dept6.nycu.edu.tw.
    15  1.24.168.192.in-addr.arpa.   IN  PTR  n2dhcp.dept6.nycu.edu.tw.
    16  1.34.168.192.in-addr.arpa.   IN  PTR  dns.dept6.nycu.edu.tw.
    17  25.34.168.192.in-addr.arpa.  IN  PTR  mail.dept6.nycu.edu.tw.
    18  80.34.168.192.in-addr.arpa.  IN  PTR  www.dept6.nycu.edu.tw.
    19  1.44.168.192.in-addr.arpa.   IN  PTR  n4dhcp.dept6.nycu.edu.tw.
    20  2.56.168.192.in-addr.arpa.  IN  PTR  dns.dept6.nycu.edu.tw.
```

Fig.P2r_3b

```
happyoyomg@Manager-node:~$ dig @127.0.0.1 www.dept6.nycu.edu.tw

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @127.0.0.1 www.dept6.nycu.edu.tw
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 56921
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 8501864734a10883010000006946cc8f20f0952b84e5218e (good)
;; QUESTION SECTION:
;www.dept6.nycu.edu.tw.         IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Sun Dec 21 00:19:27 CST 2025
;; MSG SIZE  rcvd: 78
```

Fig.P2r_4a

```
happyoyomg@Manager-node:~$ dig @127.0.0.1 -x 192.168.34.25

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @127.0.0.1 -x 192.168.34.25
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 29779
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: c3d3dcd5893c4e2e010000006946cd38ebd2be0135e3fd11 (good)
;; QUESTION SECTION:
;25.34.168.192.in-addr.arpa.    IN      PTR

;; Query time: 1 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Sun Dec 21 00:22:16 CST 2025
;; MSG SIZE  rcvd: 83
```

Fig.P2r_4b